

B.Tech. Degree II Semester Regular Examination July 2024**EE/EC/CS/IT 23-200-0206B ENVIRONMENTAL AND LIFE SCIENCES**

(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Identify the global environmental issues.

CO2: Examine the types of pollution in society along with their sources.

CO3: Elucidate the basic biological concepts via relevant industrial applications and case studies.

CO4: Evaluate the principles of design and development for exploring novel bioengineering projects.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,

L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

(5 × 2 = 10)

	Marks	BL	CO	PO
I. (a) Explain the primary production and secondary production of an ecosystem.	2	L1	1	2
(b) What is a biodiversity hotspot?	2	L2	3	2
(c) Write a short note on the impact of green revolution on the environment.	2	L2	2	2
(d) What are the various human blood substitutes?	2	L4	4	2
(e) Explain in short, the working of ECG.	2	L3	4	2

PART B

(4 × 10 = 40)

II.	Discuss the various mining techniques and their impact on the environment.	10	L1	1	2
OR					
III.	Explain the role of a forest ecosystem.	10	L1	1	2
IV.	Explain the various air pollution control techniques.	10	L2	2	2
OR					
V.	Write a short note on				
(a)	Solid waste management	5	L2	2	2
(b)	Disaster management of an earthquake.	5	L2	2	2
VI.	Mention the application of				
(a)	PHA and PLA based water filters.	6	L3	3	2
(b)	Lignolytic enzymes.	4	L3	3	2
OR					
VII.	Write a short note on				
(a)	Lotus leaf effect.	5	L3	3	2
(b)	Robotic arms for prosthetics.	5	L3	3	2
VIII.	Explain the working of				
(a)	Dialysis unit.	5	L4	4	2
(b)	Ventilator.	5	L4	4	2
OR					
IX.	Discuss the human brain function as a CPU system.	10	L4	4	2

Bloom's Taxonomy Levels

L1 – 18%, L2 – 29%, L3 – 29%, L4 – 24%.